

Code: 1BCS305**Credits: 3****L: T:P:TW&SL – 28:28:0:34****SEE Hours: 3****Course: Object Oriented Programming with Java****CIE: 50% Marks****SEE: 50% Marks****Total Marks: 100**

Prerequisites if any	C programming
Learning objectives	1. To understand and compare the principles of Object-Oriented Programming with Procedure-Oriented Programming. 2. To develop programming skills using Java and utilize its features for building software applications.

Course Outcomes:

COs	Course Outcome	Wks	SDGs
CO1	Explain Java programming fundamentals and object-oriented concepts for software development.	2, 3, 4,6	4
CO2	Apply classes, objects, methods, inheritance, interfaces, and packages to develop Java applications.	2, 3, 4, 6,9	4, 9
CO3	Develop Java programs using exception handling and object-oriented programming techniques for designing efficient and robust concurrent Java applications.	3, 4, 6, 8,9	9, 12
CO4	Evaluate Java project implementations by analysing the application of object-oriented principles in addressing real-world problems.	4, 6, 7, 8, 9	8,9,17

On the successful completion of the course, the student will be able to

Mapping with POs and PSOs:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11		PSO1	PSO2
CO1	3	2	2	1	3	2	-	2	3	2	3		2	2
CO2	3	3	3	2	3	2	2	2	3	2	3		2	2
CO3	3	3	3	2	3	2	2	2	3	2	3		3	3
CO4	3	3	3	2	3	2	2	3	3	2	3		3	3

Mapping Strength: Strong– 3 Medium – 2 Low – 1

Course Structure

Sl. No.	Modules	No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
Module – 1 Overview of Java				
1.1	Introduction to Java and OOP Principles ,JDK, JVM, Bytecode, simple java program	2	2	-
1.2	Data Types, Variables, and Arrays: Type Conversion and Casting, Automatic Type Promotion in Expressions, Arrays, Operators	2	2	-
1.3	Control Statements: Java's Selection Statements (if, The Traditional switch), Iteration Statements (while, do- while, for, The For-Each Version of the for Loop), Jump Statements (Using break, Using continue, return).	2	2	-
Module – 2: Classes and Objects				
2.1	Introducing Classes: Class Fundamentals, Declaring Objects, Assigning Object Reference Variables	1	1	-
2.2	Introducing Methods, Constructors, This Keyword, Garbage Collection.	2	2	-
2.3	Methods and Classes: Overloading Methods, Objects as Parameters, Argument Passing, Returning Objects, Access Control	2	2	-
2.4	Understanding static keyword, introducing final, Introducing super keyword.	1	1	
Module – 3: Inheritance & Interfaces				
3.1	Inheritance: Inheritance Basics, Creating a Multilevel Hierarchy.	1	1	-
3.2	When Constructors Are Executed, Method Overriding	1	1	
3.3	Super keyword , Using Abstract Classes	1	1	-
3.4	Interfaces: Interfaces, Default Interface Methods, Use static Methods in an Interface, Private Interface Methods .	2	2	-

Module – 4: Packages & Exceptions				
4.1	Packages: Packages, Packages and Member Access, Importing Packages.	2	1	-
4.2	Exceptions: Exception-Handling Fundamentals, Exception Types, Uncaught Exceptions	1	1	-
4.3	Using try and catch, Multiple catch Clauses, Nested try Statements,	1	2	-
4.4	Throw, throws, finally, Java's Built-in Exceptions, Creating Your Own Exception Subclasses.	2	2	-
Module – 5: Threads				
5.1	Multithreaded Programming: The Java Thread Model, The Main Thread, Creating a Thread, Creating Multiple Threads,	2	2	-
5.2	Using is Alive() and join(), Thread Priorities	1	1	-
5.3	Synchronization.	2	2	-
Total No. of Lecture Hours		28	-	
Total No. of Tutorial Hours			28	-
Total No. of Practical Hours				00

Textbook:

1. Herbert Schildt, *Java The Complete Reference-Eleventh Edition*, McGraw Hill; 11th Edition.

Reference Book:

1. Dr. R. Nageswara Rao, Core Java, An Integrated Approach, Dream tech Press, 2016.
2. Mahesh Bhawe and Sunil Patekar, "Programming with Java", First Edition, Pearson Education, 2008
3. Rajkumar Buyya, S Thamarai selvi, xing chenchu, Object oriented Programming with java, Tata McGraw Hill Education Private Limited, 2009
4. Richard A Johnson, An Introduction to Java Programming and Object-Oriented Application Development, Delmar Cengage Learning, 2007.
5. Michael Blaha, James Rumbaugh: Object-Oriented Modeling and Design with UML, 2nd Edition, Pearson Education/ PHI, 2007.

Online Resources:

1. Udacity Free Course

<https://www.udacity.com/course/java-programming-basics--ud282>

- Introduction To Programming In Java (by Evan Jones, Adam Marcus and Eugene Wu):
- <https://ocw.mit.edu/courses/6-092-introduction-to-programming-in-java-january-iap-2010/>
- Java Tutorial: <https://www.w3schools.com/java/>
- Java Tutorial: <https://www.javatpoint.com/java-tutorial>
- Java Tutorial: <https://www.geeksforgeeks.org/jav>